

Multiple Choice

1. **Answer:** A

Options: A) 0.7%; B) 3%; C) 0.1%; D) 1.5%; E) 1%

Note: Correct option: A - 0.7%

2. **Answer:** A

Options: A) 3.0×10^{-32} ; B) 5.5×10^{-33} ; C) 1.2×10^{-30} ; D) 8.9×10^{-36} ; E) 2.3×10^{-30}

Note: Correct option: A - 3.0×10^{-32}

3. **Answer:** A

Options: A) 1200 lb; B) 1800 lb; C) 2000 lb; D) 1000 lb; E) 2400 lb

Note: Correct option: A - 1200 lb

4. **Answer:** E

Options: A) more fragile; B) shorter; C) heavier; D) equal in height; E) taller

Note: Correct option: E - taller

5. **Answer:** A

Options: A) asteroids would orbit with a period that is three times that of Jupiter; B) asteroids would orbit with a period three times that of Mars; C) asteroids would orbit with a period twice that of Mars; D) asteroids would orbit with a period twice that of Jupiter; E) A and B

Note: Correct option: A - asteroids would orbit with a period that is three times that of Jupiter

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '36 km'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '150 million km'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'blue'.

10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '2 h'.

Long Form Response

11. **Answer:** $2.2 \times 10^{14} \text{ sec}^{-1}$. *Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.*
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
11. **Answer:** $6.5 \times 10^9 \text{ Hz}$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key